Gesture Recognition System

This project is a **Hand Gesture Recognition System** using **MediaPipe, OpenCV, and Machine Learning**. It allows users to control a slideshow using predefined hand gestures such as:

- **Open Hand** → Start the slideshow
- **Next Gesture** → Move to the next slide
- **Previous Gesture** → Move to the previous slide
- **Stop Gesture** → Exit the slideshow

Features

- Real-time hand gesture detection
- Uses **Random Forest Classifier** for recognition
- Supports dataset collection, training, and gesture prediction
- Works with a webcam for real-time interaction

Installation

1. Clone the Repository

```
git clone https://github.com/your-username/gesture-recognition.git
cd gesture-recognition
```

2. Install Dependencies

```
pip install -r requirements.txt
```

Usage

Step 1: Collect Gesture Data

Run the script to collect your hand gesture data:

```
python collect_gesture_data.py
```

- Perform gestures while pressing the corresponding keys.
- Data is saved in `gesture_data.pkl`.

Step 2: Train the Model

```
python train_gesture_model.py
```

- This trains the **Random Forest model** on collected gesture data.

- A trained model (`gesture_classifier.pkl`) is saved.

Step 3: Recognize Gestures

```
python recognize_gestures.py
```

- Ensure your trained model (`gesture_classifier.pkl`) is in the same directory.
- The system will recognize gestures in real-time and trigger slideshow actions.

File Structure

```
├── collect_gesture_data.py  # Collects gesture data
├── train_gesture_model.py   # Trains the classifier
├── recognize_gestures.py    # Runs the recognition system
├── gesture_data/            # Folder containing gesture dataset
├── gesture_data.pkl         # Saved gesture data
├── gesture_classifier.pkl   # Trained model
├── requirements.txt         # Required Python packages
├── README.md                # Project documentation
```

Requirements

- Python 3.12
- OpenCV
- MediaPipe
- scikit-learn
- NumPy
- PyAutoGUI (for slideshow control)

Future Improvements

- Add **deep learning models** (CNNs) for improved accuracy.
- Expand the number of gestures.
- Enhance the UI for better usability.

License

This project is open-source under the MIT License.